Probability you can argue about

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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Week 2 • Chapter 2, Introduction to Probability

How to use this sheet

Try every question before the tutorial, in your group if you like, and bring your working on paper. Several of these questions have no single correct answer: they ask you to take a position and defend it with the numbers. A one line answer with no reasoning earns nothing, and a wrong answer argued carefully is worth more than a right answer copied from a friend.

Q1. A blast round uses detonators that fire independently. Each one misfires with probability 0.03.

(a) Two holes are charged. Write out the sample space, and the events "at least one misfires" and "both fire". Are those two events complements?

(b) Find the probability that at least one of 5 holes misfires, and that at least one of 20 holes misfires.

(c) A supervisor says "a 3% misfire rate is nothing to worry about". Use your answers to write the two sentences you would say back to him.

(d) Where is the independence assumption most likely to fail on a real bench? If it fails, does your answer to (b) go up or down?

Q2. For two events A and B, a report states $P(A) = 0.6$ and $P(B) = 0.5$. Decide whether each further claim is possible, and justify it from the axioms.

(a) $P(A \cap B) = 0.05$.

(b) $P(A \cap B) = 0.2$, and therefore $P(A \cup B) = 0.9$.

(c) A and B are mutually exclusive.

(d) A and B are independent. If so, what must $P(A \cap B)$ be?

(e) Find the smallest and largest values $P(A \cap B)$ could take.

Q3. Four hundred concrete cubes from two suppliers were crushed at 28 days.

	Passed	Failed	Total
Supplier A	168	12	180
Supplier B	196	24	220
Total	364	36	400

A cube is picked at random from the 400.

(a) Find $P(\text{fail})$, $P(\text{fail} \mid A)$ and $P(\text{fail} \mid B)$.

(b) Are "fails" and "comes from supplier A" independent? Show the check.

(c) The site engineer concludes "supplier B is worse, stop buying from them". The failure rates differ by about four percentage points on 400 cubes. What would you need to know before acting on that conclusion?

(d) Supplier B points out that they supplied more cubes, so of course they have more failures. Is that a fair defence?

Q4. A vibration sensor on a gearbox raises an alarm. Historically 2% of gearboxes are genuinely developing a fault at any time. The sensor raises the alarm for 95% of faulty gearboxes, and also for 10% of healthy ones.

(a) The alarm has just gone off. What is the probability the gearbox is really faulty?

(b) Explain, to a maintenance crew with no statistics, why that number is so far below 95%.

(c) The vendor offers two upgrades: raise the detection rate to 99%, or cut the false alarm rate to 1%. Which buys more, and by how much?

(d) The crew start ignoring the alarm. Is that irrational? Answer with probabilities, then say what you would change about the system.

(e) On a fleet of older gearboxes where 10% are genuinely faulty, recompute (a). What does the comparison tell you about using the same alarm everywhere?

Q5. Two events each have positive probability.

(a) Show that if they are mutually exclusive, they cannot be independent.

(b) Give an engineering example of two events that are mutually exclusive, and one of two events that are independent.

(c) A student writes "A and B are unrelated, so $P(A \cap B) = 0$". Diagnose the error in one sentence.

Q6. Ten boreholes have been drilled. Exactly two of them hit high grade ore, though nobody knows which yet. Four holes are chosen at random for assay.

(a) In how many ways can the four be chosen?

(b) Find the probability that both high grade holes are among the four.

(c) Find the probability that neither of them is.

(d) The geologist does not choose at random: he picks the four holes nearest the old shaft. Do your answers still describe his chance of finding the ore? Explain what has changed.

Q7. A dewatering system uses pumps that each work with probability 0.9, independently.

(a) Two pumps are installed in parallel, so the system works if either works. Find the system reliability.

(b) Three components in series, each working with probability 0.95, must all work. Find the system reliability, and comment on what adding components in series does.

(c) How many parallel pumps are needed for a system reliability of at least 0.999?

(d) Both pumps in (a) are fed from one power line that fails with probability 0.04. Recompute the reliability. Which is the weak point now, and what does this say about the word "independent"?

Q8. A workshop buys bearings from three suppliers: A supplies 50% with a 2% defect rate, B supplies 30% with 4%, and C supplies 20% with 7%.

(a) What proportion of all bearings in the store are defective?

(b) A defective bearing is found. Which supplier did it most probably come from?

(c) Supplier C provides only one bearing in five. What share of the defects do they account for? Explain the gap in plain words.

(d) The manager decides to drop supplier C. Before he does, what else should he weigh up, and what statistic would you ask for?

Q9. Each of these sentences confuses $P(A \mid B)$ with $P(B \mid A)$. Write out both conditional probabilities in words for each one, and say which the speaker actually has evidence for.

(a) "Most machines that fail were overdue for service, so a machine that is overdue will probably fail."

(b) "Nearly every flooded pit we have seen followed heavy rain, so heavy rain nearly always floods the pit."

(c) "Ninety nine percent of students who failed had poor attendance, so poor attendance means you will almost certainly fail."

Q10. Design a small study, on paper, to estimate the probability that the campus network drops at least once during a one hour lecture.

(a) What is the experiment, and what is the sample space?

(b) How would you collect the data, and over how many lectures?

(c) Name two things that would bias your estimate, and say which way.

(d) Your estimate comes out as 0.18. What exactly does that number mean, and what does it not mean about next Tuesday's lecture?

Several of these questions are designed so that the obvious answer is wrong. If an answer feels surprising, check the arithmetic, and if the arithmetic holds, believe it and be ready to defend it in the tutorial.